

Snidhi Reddy Gudoor

Data Engineer

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PROFESSIONAL SUMMARY

- Data Engineer with 5+ years of experience** designing, optimizing, and scaling **data platforms, pipelines, and ML infrastructure** across **finance, tech, and industrial IoT domains**. Proven expertise in building **real-time and batch data systems** using **Databricks, Snowflake, Apache Spark, Kafka, Flink, and Airflow**. Skilled in developing **end-to-end machine learning pipelines** and **data transformation frameworks** that improve model accuracy, reduce latency, and enhance data reliability.
- Hands-on experience with **AWS, Azure, and modern data lakehouse architectures (Delta Lake, Iceberg)**, delivering measurable gains in **ETL performance, cost optimization, and data quality**. Adept at implementing **CI/CD pipelines, dbt-based transformations, and MLOps workflows** using **MLflow and Azure ML**. Passionate about leveraging **Generative AI (Azure OpenAI)** and **advanced analytics** to drive **business intelligence and data-driven decision-making**.

TECHNICAL SKILLS

Programming & Scripting: Python, Java, Scala, R, Shell Scripting, SQL (T-SQL, PL/SQL), NoSQL (MongoDB, DynamoDB, Cassandra)
Big Data & Streaming: Apache Spark, Hadoop, Apache Kafka, Kinesis, Azure Event Hubs, Data Lakehouse Architecture (Delta Lake, Iceberg)
ETL & Data Orchestration: Apache Airflow, AWS Glue, dbt Core, Prefect, SSIS, Batch Processing, Pipeline Automation, Data Integration
Databases & Data Warehousing: Snowflake, Redshift, PostgreSQL, Oracle, MySQL, Data Modeling, Governance, Query Optimization
Cloud Platforms: AWS (EC2, S3, Lambda, Glue, Athena, EMR, Redshift), Azure (Data Factory, Databricks, Azure ML, Synapse, OpenAI)
DevOps & Infrastructure as Code: Docker, Kubernetes (EKS, AKS), Jenkins, Git, GitHub, GitLab CI/CD, Terraform
Machine Learning & MLOps: Scikit-learn, Pandas, NumPy, MLflow, Model Deployment, Model Monitoring, Feature Store, NLP, LLM Pipelines
Business Intelligence & Visualization: Power BI, Tableau, SSRS, Advanced Excel (Formulas, Pivot Tables), KPI Dashboard Development
Other Core Competencies: Agile / Scrum Methodology, SDLC, Great Expectations, UDF Development, Recursive CTEs, Performance Tuning

CERTIFICATION

- Databricks Certified Data Engineer Associate**

EXPERIENCE

BlackRock – Senior Data Engineer

Location: New York, USA

Duration: January 2025 – Present

- Developed and productionized **end-to-end ML pipelines** in **Python** using **scikit-learn** and **PySpark MLlib** for **trade surveillance and liquidity forecasting**, improving **model accuracy by 18%** and reducing **false positives by 10%**.
- Built modular **data transformation pipelines** using **dbt Core, Snowflake, and SQL**, creating reusable, test-driven models that standardized **feature engineering** logic and improved data consistency for **machine learning workflows by 15%**.
- Optimized **large-scale financial data pipelines** in **Databricks (Delta Lake)** using **Z-Ordering, advanced partitioning, and cluster auto-scaling**, improving **ETL throughput by 25%** and reducing **cloud storage costs by 15%** across **multi-asset datasets**.
- Collaborated with **Quantitative Research** and **Aladdin platform teams** to deploy **machine learning models** using **Databricks, MLflow, and Azure ML**, ensuring **data integrity** for **20+ production ML models** and reducing **training-serving skew by 40%**.
- Designed and deployed **Generative AI (GenAI) solutions** using **Azure OpenAI**, automating **financial narrative generation**, and built **interactive dashboards** in **Power BI** connected to **Azure Synapse Analytics**, cutting **reporting cycles by 30%**.

Airbnb – Data Engineer

Location: New York, USA

Duration: April 2024 – January 2025

- Designed and implemented **low-latency data pipelines** using **Apache Kafka, Apache Flink, and AWS**, processing **billions of daily events** (searches, bookings, pricing changes) to enable **real-time analytics** for **pricing trends and market saturation metrics**.
- Managed and optimized **Airbnb Data Lake** performance on **Amazon S3** using **Apache Iceberg** and **Apache Spark** (compaction, partition tuning), reducing **analyst query times by 40%** and improving **data accessibility**.
- Developed and deployed **12+ production DAGs** in **Apache Airflow** to automate **complex ELT workflows**, incorporating **custom operators, SLA monitoring, and retry logic**, achieving **97% pipeline reliability**.
- Engineered **data quality and validation frameworks** in **Python (Pandas)** and **PySpark** for **distributed data validation**, improving **data integrity** and reducing **pipeline failure rates by 15%**.
- Implemented a **CI/CD pipeline** using **GitHub Actions** to automate testing and deployment of **dbt models** and **PySpark scripts**, cutting **manual release errors by 20%** and increasing **deployment frequency**.

General Electric – Data Engineer

Location: India

Duration: June 2019 – November 2022

- Architected and implemented a **real-time IoT data processing platform** using **Apache Kafka** and **Spark Streaming** to ingest and analyze **high-velocity sensor data**, enabling **anomaly detection** and reducing **turbine maintenance response time by 20%**.
- Engineered and optimized **ETL pipelines** in **Python** and **SQL** on **AWS (Glue, Lambda, S3)** to process **terabytes of daily sensor telemetry data** from **industrial turbines**, powering **predictive maintenance models** that reduced **unplanned downtime**.
- Improved **data storage efficiency** by implementing **columnar file formats (Parquet)** and **intelligent partitioning strategies** in **AWS S3**, cutting **query execution times by 25%** and reducing **storage costs**.
- Utilized **Apache Airflow** to orchestrate **multi-source data workflows**, automating **data ingestion and transformation** from **ERP (SAP), CRM (Salesforce), and IoT data streams**, saving **10+ hours of manual effort per week**.

EDUCATION

Master's in Computer Science, University of Central Missouri, Missouri, USA.

Bachelor's in Computer Science, St. Martin's Engineering College, Telangana, India.